Matrices (AI HL)

Section 1: Matrix Multiplication and Inverses

1 Compute the matrix product AB for $A = [[1, 2], [3, 4]]$ and $B = [[2, 0], [1, 3]]$.

a. $[[4, 6], [10, 12]]$

b. $[[2, 0], [3, 12]]$

c. $[[4, 10], [6, 12]]$

d. $[[3, 2], [4, 7]]$

2 Find the inverse of the matrix $M = [[3, 5], [1, 2]]$.

a. $[[2, -5], [-1, 3]]$

b. $[[-2, 5], [1, -3]]$

c. $[[3, -1], [-5, 2]]$

d. $[[1/3, 1/5], [1, 1/2]]$

3 If A and B are invertible matrices of the same dimension, what is $(AB)^{-1}$?

a. $B^{-1}A^{-1}$

b. $A^{-1}B^{-1}$

c. BA^{-1}

d. AB^{-1}

Section 2: Determinants and Singularity

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4 For what value of k is the matrix $A = [[k, 6], [2, 3]]$ singular (non-invertible)?

a. $k = 4$

b. $k = 6$

c. $k = 0$

d. $k = 9$

5 If A is a 3×3 matrix with $\det(A) = 5$, what is $\det(2A)$?

a. 40

b. 10

c. 20

d. 80

6 If $\det(A) = 4$, what is $\det(A^{-1})$?

a. $\frac{1}{4}$

b. -4

c. 4

d. 16

Section 3: Eigenvalues and Eigenvectors

7 Find the eigenvalues of the matrix $A = [[4, 2], [1, 3]]$.

a. $\lambda = 5$ and $\lambda = 2$

b. $\lambda = 4$ and $\lambda = 3$

c. $\lambda = 6$ and $\lambda = 1$

d. $\lambda = 7$ and $\lambda = 0$

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8 Find an eigenvector corresponding to $\lambda = 5$ for $A = [[4, 2], [1, 3]]$.

a. $[2, 1]^T$

b. $[1, 2]^T$

c. $[1, -1]^T$

d. $[2, -1]^T$

9 What is the sum of the eigenvalues of an $n \times n$ matrix A equal to?

a. The trace of A (sum of diagonal entries)

b. The determinant of A

c. The rank of A

d. n

Section 4: Diagonalization & Matrix Powers

10 If a matrix A is diagonalized as $A = PDP^{-1}$, how is the power A^n computed efficiently?

a. $A^n = PD^nP^{-1}$

b. $A^n = P^nD^n(P^{-1})^n$

c. $A^n = D^n$

d. $A^n = PDP^{-1}n$

11 If $D = [[2, 0], [0, 3]]$, what is D^4?

a. $[[16, 0], [0, 81]]$

b. $[[8, 0], [0, 12]]$

c. $[[24, 0], [0, 24]]$

d. $[[16, 81], [0, 0]]$

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12 When is an $n \times n$ matrix A guaranteed to be diagonalizable?

a. When it has n linearly independent eigenvectors (or n distinct eigenvalues)

b. When its determinant is non-zero

c. Only if it is a symmetric identity matrix

d. When all its entries are positive

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